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Statistics

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	123 questions
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MCQ (Single Correct Answer) 123 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A set of four observations has mean 1 and variance 13. Another set of six observations has mean 2 and variance 1 . Then, the variance of all these 10 observations is equal to :

(A) 5.96 (B) 6.14 (C) 6.04 (D) 6.24

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Let the mean and the variance of seven observations $2, 4, \alpha, 8, \beta, 12, 14, \alpha < \beta$, be 8 and 16 respectively. Then the quadratic equation whose roots are $3\alpha + 2$ and $2\beta + 1$ is :

(A) $x^2 - 35x + 306 = 0$ (B) $x^2 - 41x + 420 = 0$

(C) $x^2 - 45x + 506 = 0$ (D) $x^2 - 37x + 342 = 0$

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

A data consists of 20 observations $x_1, x_2, \ldots, x_{20}$. If $\sum_{i=1}^{20} (x_i + 5)^2 = 2500$ and $\sum_{i=1}^{20} (x_i - 5)^2 = 100$, then the ratio of mean to standard deviation of this data is :

(A) 2 : 1 (B) 3 : 1 (C) 3 : 2 (D) 4 : 1

Open solution

Q4. JEE Main 2026 (Online) 5th April Evening Shift

A variable X takes values $0, 0, 2, 6, 12, 20, \ldots, n(n-1)$ with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3, {}^nC_4, {}^nC_5, \ldots, {}^nC_n$, respectively. If the mean of this data is 60 , then its median is :

(A) 56 (B) 42 (C) 72 (D) 90

Open solution

Q5. JEE Main 2026 (Online) 5th April Morning Shift

The mean deviation about the mean for the data

x_i	5	7	9	10	12	15
f_i	8	6	2	2	2	6

is equal to:

(A) $\frac{40}{13}$ (B) $\frac{42}{13}$ (C) $\frac{44}{13}$ (D) $\frac{46}{13}$

Open solution

Q6. JEE Main 2026 (Online) 4th April Evening Shift

For 10 observations $x_1, x_2, \ldots, x_{10}$, if $\sum_{i=1}^{10} (x_i + 2)^2 = 180$ and $\sum_{i=1}^{10} (x_i - 1)^2 = 90$, then their standard deviation is :

(A) 2 (B) $\sqrt{3}$ (C) $2\sqrt{2}$ (D) 3

Open solution

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Q7. JEE Main 2026 (Online) 4th April Morning Shift

Suppose that the mean and median of the non-negative numbers $21, 8, 17, a, 51, 103, b, 13, 67, (a > b)$, are 40 and 21 , respectively. If the mean deviation about the median is 26 , then $2a$ is equal to :

Open solution

(A) 109 (B) 117 (C) 161 (D) 131

Q8. JEE Main 2026 (Online) 2nd April Evening Shift

The mean and variance of **n** observations are 8 and 16, respectively. If the sum of the first (n − 1) observations is 48 and the sum of squares of the first (n − 1) observations is 496, then the value of n is :

Open solution

(A) 21 (B) 16 (C) 13 (D) 7

Q9. JEE Main 2026 (Online) 2nd April Morning Shift

If the mean of the data

Class	5-10	10-15	15-20	20-25	25-30	30-35
Frequency	2	k	28	54	k + 1	5

is 21, then k is one of the roots of the equation:

Open solution

(A) $2x^2 - 23x - 10 = 0$ (B) $4x^2 - 35x + 24 = 0$

(C) $2x^2 - 19x - 10 = 0$ (D) $2x^2 - 35x + 98 = 0$

Q10. JEE Main 2026 (Online) 28th January Morning Shift

The mean and variance of 10 observations are 9 and 34.2 , respectively. If 8 of these observations are $2, 3, 5, 10, 11, 13, 15, 21$, then the mean deviation about the median of all the 10 observations is

Open solution

(A) 7 (B) 4 (C) 5 (D) 6

Q11. JEE Main 2026 (Online) 24th January Evening Shift

Let $\mathrm{X} = \{x \in \mathrm{N} : 1 \leq x \leq 19\}$ and for some $a, b \in \mathbb{R}, \mathrm{Y} = \{ax + b : x \in \mathrm{X}\}$. If the mean and variance of the elements of Y are 30 and 750 , respectively, then the sum of all possible values of b is

Open solution

(A) 20 (B) 100 (C) 80 (D) 60

Q12. JEE Main 2026 (Online) 24th January Morning Shift

The mean and variance of a data of 10 observations are 10 and 2 , respectively. If an observations α in this data is replaced by β, then the mean and variance become 10.1 and 1.99 , respectively. Then $\alpha + \beta$ equals

Open solution

(A) 15 (B) 10 (C) 5 (D) 20

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Q13. JEE Main 2026 (Online) 23rd January Evening Shift

If the mean and the variance of the data

Class	$4-8$	$8-12$	$12-16$	$16-20$
Frequency	3	λ	4	7

are μ and 19 respectively, then the value of $\lambda+\mu$ is :

(A) 21 (B) 19 (C) 18 (D) 20

Open solution

Q14. JEE Main 2026 (Online) 23rd January Morning Shift

Let the mean and variance of 8 numbers $-10,-7,-1,x,y,9,2,16$ be $\frac{7}{2}$ and $\frac{293}{4}$, respectively.

Then the mean of 4 numbers $x,y,x+y+1,|x-y|$ is :

(A) 11 (B) 9 (C) 10 (D) 12

Open solution

Q15. JEE Main 2026 (Online) 22nd January Evening Shift

If the mean deviation about the median of the numbers $\mathrm{k},2\mathrm{k},3\mathrm{k},\ldots,1000\mathrm{k}$ is 500 , then k^2 is equal to :

(A) 1 (B) 16 (C) 9 (D) 4

Open solution

Q16. JEE Main 2025 (Online) 7th April Morning Shift

The mean and standard deviation of 100 observations are 40 and 5.1 , respectively. By mistake one observation is taken as 50 instead of 40 . If the correct mean and the correct standard deviation are μ and σ respectively, then $10(\mu+\sigma)$ is equal to

(A) 447 (B) 445 (C) 449 (D) 451

Open solution

Q17. JEE Main 2025 (Online) 4th April Evening Shift

Let the mean and the standard deviation of the observation $2,3,3,4,5,7,a,b$ be 4 and $\sqrt{2}$ respectively. Then the mean deviation about the mode of these observations is :

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) 1 (D) 2

Open solution

Q18. JEE Main 2025 (Online) 3rd April Evening Shift

Let the Mean and Variance of five observations $x_1=1,x_2=3,x_3=a,x_4=7$ and $x_5=\mathrm{b},a>\mathrm{b}$, be 5 and 10 respectively. Then the Variance of the observations $n+x_n,n=1,2,\ldots,5$ is

(A) 17 (B) 16 (C) 16.4 (D) 17.4

Open solution

Q19. JEE Main 2025 (Online) 2nd April Evening Shift

If the mean and the variance of $6,4,a,8,b,12,10,13$ are 9 and 9.25 respectively, then $a+b+ab$ is equal to :

(A) 103 (B) 106 (C) 100 (D) 105

Open solution

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Q20. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

Let $x_1, x_2, \ldots, x_{10}$ be ten observations such that $\sum_{i=1}^{10}(x_i - 2) = 30$, $\sum_{i=1}^{10}(x_i - \beta)^2 = 98$, $\beta > 2$, and their variance is $\frac{4}{5}$. If μ and σ^2 are respectively the mean and the variance of $2(x_1 - 1) + 4\beta, 2(x_2 - 1) + 4\beta, \ldots, 2(x_{10} - 1) + 4\beta$, then $\frac{\beta\mu}{\sigma^2}$ is equal to :

(A) 100 (B) 90 (C) 120 (D) 110

Q21. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

For a statistical data $x_1, x_2, \ldots, x_{10}$ of 10 values, a student obtained the mean as 5.5 and $\sum_{i=1}^{10} x_i^2 = 371$. He later found that he had noted two values in the data incorrectly as 4 and 5 , instead of the correct values 6 and 8 , respectively. The variance of the corrected data is

(A) 5 (B) 7 (C) 9 (D) 4

Q22. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Marks obtains by all the students of class 12 are presented in a freqency distribution with classes of equal width. Let the median of this grouped data be 14 with median class interval 12-18 and median class frequency 12. If the number of students whose marks are less than 12 is 18 , then the total number of students is :

(A) 52 (B) 44 (C) 40 (D) 48

Q23. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

If the variance of the frequency distribution

x	c	$2c$	$3c$	$4c$	$5c$	$6c$
f	2	1	1	1	1	1

is 160, then the value of $c \in N$ is

(A) 5 (B) 8 (C) 6 (D) 7

Q24. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

The frequency distribution of the age of students in a class of 40 students is given below.

Age	**15**	**16**	**17**	**18**	**19**	**20**
No of Students	5	8	5	12	x	y

If the mean deviation about the median is 1.25, then $4x + 5y$ is equal to :

(A) 43 (B) 46 (C) 44 (D) 47

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Q25. JEE Main 2024 (Online) 6th April Morning Shift

The mean and standard deviation of 20 observations are found to be 10 and 2 , respectively. On rechecking, it was found that an observation by mistake was taken 8 instead of 12. The correct standard deviation is

Open solution

(A) 1.94 (B) $\sqrt{3.96}$ (C) $\sqrt{3.86}$ (D) 1.8

Q26. JEE Main 2024 (Online) 4th April Morning Shift

Let $\alpha, \beta \in \mathbf{R}$. Let the mean and the variance of 6 observations $-3, 4, 7, -6, \alpha, \beta$ be 2 and 23, respectively. The mean deviation about the mean of these 6 observations is :

Open solution

(A) $\frac{16}{3}$ (B) $\frac{11}{3}$ (C) $\frac{14}{3}$ (D) $\frac{13}{3}$

Q27. JEE Main 2024 (Online) 1st February Evening Shift

Consider 10 observations $x_1, x_2, \ldots, x_{10}$ such that $\sum_{i=1}^{10}(x_i - \alpha) = 2$ and $\sum_{i=1}^{10}(x_i - \beta)^2 = 40$, where α, β are positive integers. Let the mean and the variance of the observations be $\frac{6}{5}$ and $\frac{84}{25}$ respectively. Then $\frac{\beta}{\alpha}$ is equal to :

Open solution

(A) 2 (B) 1 (C) $\frac{5}{2}$ (D) $\frac{3}{2}$

Q28. JEE Main 2024 (Online) 1st February Morning Shift

Let the median and the mean deviation about the median of 7 observations $170, 125, 230, 190, 210$, a, b be 170 and $\frac{205}{7}$ respectively. Then the mean deviation about the mean of these 7 observations is :

Open solution

(A) 31 (B) 28 (C) 30 (D) 32

Q29. JEE Main 2024 (Online) 31st January Evening Shift

Let the mean and the variance of 6 observations $a, b, 68, 44, 48, 60$ be 55 and 194, respectively. If $a > b$, then $a + 3b$ is

Open solution

(A) 180 (B) 210 (C) 190 (D) 200

Q30. JEE Main 2024 (Online) 30th January Morning Shift

Let M denote the median of the following frequency distribution

Open solution

Class	0 - 4	4 - 8	8 - 12	12 - 16	16 - 20
Frequency	3	9	10	8	6

Then 20M is equal to :

(A) 104 (B) 52 (C) 208 (D) 416

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Q31. JEE Main 2024 (Online) 29th January Evening Shift

If the mean and variance of five observations are $\frac{24}{5}$ and $\frac{194}{25}$ respectively and the mean of the first four observations is $\frac{7}{2}$, then the variance of the first four observations is equal to

(A) $\frac{5}{4}$ (B) $\frac{4}{5}$ (C) $\frac{105}{4}$ (D) $\frac{77}{12}$

Open solution

Q32. JEE Main 2024 (Online) 27th January Morning Shift

Let $a_1, a_2, \ldots a_{10}$ be 10 observations such that $\sum_{k=1}^{10} a_k = 50$ and $\sum_{\forall k<j} a_k \cdot a_j = 1100$. Then the standard deviation of $a_1, a_2, \ldots, a_{10}$ is equal to :

(A) 5 (B) $\sqrt{115}$ (C) 10 (D) $\sqrt{5}$

Open solution

Q33. JEE Main 2023 (Online) 15th April Morning Shift

The mean and standard deviation of 10 observations are 20 and 8 respectively. Later on, it was observed that one observation was recorded as 50 instead of 40. Then the correct variance is :

(A) 11 (B) 12 (C) 13 (D) 14

Open solution

Q34. JEE Main 2023 (Online) 11th April Evening Shift

Let the mean of 6 observations $1, 2, 4, 5, x$ and y be 5 and their variance be 10 . Then their mean deviation about the mean is equal to :

(A) $\frac{10}{3}$ (B) $\frac{8}{3}$ (C) $\frac{7}{3}$ (D) 3

Open solution

Q35. JEE Main 2023 (Online) 11th April Morning Shift

Let sets A and B have 5 elements each. Let the mean of the elements in sets A and B be 5 and 8 respectively and the variance of the elements in sets A and B be 12 and 20 respectively. A new set C of 10 elements is formed by subtracting 3 from each element of A and adding 2 to each element of B. Then the sum of the mean and variance of the elements of C is _____.

(A) 36 (B) 40 (C) 38 (D) 32

Open solution

Q36. JEE Main 2023 (Online) 10th April Evening Shift

Let μ be the mean and σ be the standard deviation of the distribution

x_i	0	1	2	3	4	5
f_i	$k+2$	$2k$	k^2-1	k^2-1	k^2+1	$k-3$

where $\sum f_i = 62$. If $[x]$ denotes the greatest integer $\leq x$, then $\left[\mu^2+\sigma^2\right]$ is equal to :

(A) 9 (B) 8 (C) 6 (D) 7

Open solution

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Q37. JEE Main 2023 (Online) 8th April Evening Shift

Let the mean and variance of 12 observations be $\frac{9}{2}$ and 4 respectively. Later on, it was observed that two observations were considered as 9 and 10 instead of 7 and 14 respectively. If the correct variance is $\frac{m}{n}$, where m and n are coprime, then $\mathrm{m}+\mathrm{n}$ is equal to :

(A) 317 (B) 316 (C) 314 (D) 315

Open solution

Q38. JEE Main 2023 (Online) 6th April Morning Shift

The mean and variance of a set of 15 numbers are 12 and 14 respectively. The mean and variance of another set of 15 numbers are 14 and σ^2 respectively. If the variance of all the 30 numbers in the two sets is 13 , then σ^2 is equal to :

(A) 12 (B) 11 (C) 10 (D) 9

Open solution

Q39. JEE Main 2023 (Online) 1st February Evening Shift

Let $9=x_1<x_2<\ldots<x_7$ be in an A.P. with common difference d. If the standard deviation of $x_1, x_2 \ldots, x_7$ is 4 and the mean is $\bar{x}$, then $\bar{x}+x_6$ is equal to :

(A) $2\left(9+\frac{8}{\sqrt{7}}\right)$ (B) 25

(C) $18\left(1+\frac{1}{\sqrt{3}}\right)$ (D) 34

Open solution

Q40. JEE Main 2023 (Online) 1st February Morning Shift

The mean and variance of 5 observations are 5 and 8 respectively. If 3 observations are 1, 3, 5, then the sum of cubes of the remaining two observations is :

(A) 1792 (B) 1216 (C) 1456 (D) 1072

Open solution

Q41. JEE Main 2023 (Online) 31st January Evening Shift

Let the mean and standard deviation of marks of class A of 100 students be respectively 40 and $\alpha(>0)$, and the mean and standard deviation of marks of class B of n students be respectively 55 and $30-\alpha$. If the mean and variance of the marks of the combined class of $100+\mathrm{n}$ students are respectively 50 and 350 , then the sum of variances of classes A and B is :

(A) 450 (B) 900 (C) 650 (D) 500

Open solution

Q42. JEE Main 2023 (Online) 30th January Evening Shift

Let S be the set of all values of a_1 for which the mean deviation about the mean of 100 consecutive positive integers $a_1, a_2, a_3, \ldots, a_{100}$ is 25 . Then S is :

(A) $\{9\}$ (B) ϕ (C) $\{99\}$ (D) N

Open solution

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Q43. JEE Main 2023 (Online) 25th January Morning Shift

The mean and variance of the marks obtained by the students in a test are 10 and 4 respectively. Later, the marks of one of the students is increased from 8 to 12. If the new mean of the marks is 10.2, then their new variance is equal to :

(A) 3.92 (B) 4.08 (C) 3.96 (D) 4.04

Open solution

Q44. JEE Main 2023 (Online) 24th January Evening Shift

Let the six numbers $a_1, a_2, a_3, a_4, a_5, a_6$, be in A.P. and $a_1 + a_3 = 10$. If the mean of these six numbers is $\frac{19}{2}$ and their variance is σ^2, then $8\sigma^2$ is equal to :

(A) 220 (B) 210 (C) 105 (D) 200

Open solution

Q45. JEE Main 2022 (Online) 30th July Evening Shift

If the mean and median of the data

x	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$	
f	3	6	2	x	y	$\sum f = 20$

are equal, then xy^2 is equal to

(A) 54 (B) 80 (C) 100 (D) 108

Open solution

Q46. JEE Main 2022 (Online) 25th July Evening Shift

If the mean deviation about median for the numbers 3, 5, 7, 2k, 12, 16, 21, 24, arranged in the ascending order, is 6 then the median is :

(A) 11.5 (B) 10.5 (C) 12 (D) 11

Open solution

Q47. JEE Main 2022 (Online) 29th June Evening Shift

The number of values of a $\in$ N such that the variance of 3, 7, 12, a, 43 $-$ a is a natural number is :

(A) 0 (B) 2 (C) 5 (D) infinite

Open solution

Q48. JEE Main 2022 (Online) 29th June Morning Shift

Let the mean and the variance of 5 observations x_1, x_2, x_3, x_4, x_5 be $\frac{24}{5}$ and $\frac{194}{25}$ respectively. If the mean and variance of the first 4 observations are $\frac{7}{2}$ and a respectively, then $(4a + x_5)$ is equal to:

(A) 13 (B) 15 (C) 17 (D) 18

Open solution

Q49. JEE Main 2022 (Online) 27th June Evening Shift

The mean and variance of the data 4, 5, 6, 6, 7, 8, x, y, where x < y, are 6 and $\frac{9}{4}$ respectively. Then $x^4 + y^2$ is equal to :

(A) 162 (B) 320 (C) 674 (D) 420

Open solution

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Q50. JEE Main 2022 (Online) 26th June Evening Shift

The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to :

(A) 10 (B) 36 (C) 43 (D) 60

Open solution

Q51. JEE Main 2022 (Online) 26th June Morning Shift

The mean of the numbers a, b, 8, 5, 10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then 25 M is equal to :

(A) 60 (B) 55 (C) 50 (D) 45

Open solution

Q52. JEE Main 2021 (Online) 31st August Evening Shift

The mean and variance of 7 observations are 8 and 16 respectively. If two observations are 6 and 8, then the variance of the remaining 5 observations is :

(A) $\frac{92}{5}$ (B) $\frac{134}{5}$ (C) $\frac{536}{25}$ (D) $\frac{112}{5}$

Open solution

Q53. JEE Main 2021 (Online) 26th August Morning Shift

The mean and standard deviation of 20 observations were calculated as 10 and 2.5 respectively. It was found that by mistake one data value was taken as 25 instead of 35. If α and $\sqrt{\beta}$ are the mean and standard deviation respectively for correct data, then (α, β) is :

(A) (11, 26) (B) (10.5, 25) (C) (11, 25) (D) (10.5, 26)

Open solution

Q54. JEE Main 2021 (Online) 27th July Evening Shift

Let the mean and variance of the frequency distribution

x :	$x_1 = 2$	$x_2 = 6$	$x_3 = 8$	$x_4 = 9$
f :	4	4	α	β

be 6 and 6.8 respectively. If x_3 is changed from 8 to 7, then the mean for the new data will be :

(A) 4 (B) 5 (C) $\frac{17}{3}$ (D) $\frac{16}{3}$

Open solution

Q55. JEE Main 2021 (Online) 27th July Morning Shift

If the mean and variance of the following data : 6, 10, 7, 13, a, 12, b, 12 are 9 and $\frac{37}{4}$

respectively, then $(a - b)^2$ is equal to :

(A) 24 (B) 12 (C) 32 (D) 16

Open solution

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Q56. JEE Main 2021 (Online) 25th July Evening Shift

The first of the two samples in a group has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, then the standard deviation of the second sample is :

(A) 8 (B) 6 (C) 4 (D) 5

Open solution

Q57. JEE Main 2021 (Online) 20th July Evening Shift

If the mean and variance of six observations 7, 10, 11, 15, a, b are 10 and $\frac{20}{3}$, respectively, then the value of | a − b | is equal to :

(A) 9 (B) 11 (C) 7 (D) 1

Open solution

Q58. JEE Main 2021 (Online) 20th July Morning Shift

The mean of 6 distinct observations is 6.5 and their variance is 10.25. If 4 out of 6 observations are 2, 4, 5 and 7, then the remaining two observations are :

(A) 10, 11 (B) 3, 18 (C) 8, 13 (D) 1, 20

Open solution

Q59. JEE Main 2021 (Online) 18th March Evening Shift

Let in a series of 2n observations, half of them are equal to a and remaining half are equal to −a. Also by adding a constant b in each of these observations, the mean and standard deviation of new set become 5 and 20, respectively. Then the value of $a^2 + b^2$ is equal to :

(A) 425 (B) 250 (C) 925 (D) 650

Open solution

Q60. JEE Main 2021 (Online) 16th March Morning Shift

Consider three observations a, b, and c such that b = a + c. If the standard deviation of a + 2, b + 2, c + 2 is d, then which of the following is true?

(A) $b^2 = 3(a^2 + c^2) + 9d^2$ (B) $b^2 = 3(a^2 + c^2) - 9d^2$

(C) $b^2 = 3(a^2 + c^2 + d^2)$ (D) $b^2 = a^2 + c^2 + 3d^2$

Open solution

Q61. JEE Main 2020 (Online) 6th September Morning Slot

If $\sum_{i=1}^{n}(x_i - a) = n$ and $\sum_{i=1}^{n}(x_i - a)^2 = na$

(n, a > 1) then the standard deviation of n observations $x_1, x_2, ..., x_n$ is :

(A) $a - 1$ (B) $n\sqrt{a-1}$ (C) $\sqrt{n(a-1)}$ (D) $\sqrt{a-1}$

Open solution

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Q62. JEE Main 2020 (Online) 5th September Evening Slot

If the mean and the standard deviation of the data 3, 5, 7, a, b are 5 and 2 respectively, then a and b are the roots of the equation :

(A) $x^2 - 20x + 18 = 0$ (B) $2x^2 - 20x + 19 = 0$

(C) $x^2 - 10x + 18 = 0$ (D) $x^2 - 10x + 19 = 0$

Open solution

Q63. JEE Main 2020 (Online) 5th September Morning Slot

The mean and variance of 7 observations are 8 and 16, respectively. If five observations are 2, 4, 10, 12, 14, then the absolute difference of the remaining two observations is :

(A) 2 (B) 3 (C) 1 (D) 4

Open solution

Q64. JEE Main 2020 (Online) 4th September Morning Slot

The mean and variance of 8 observations are 10 and 13.5, respectively. If 6 of these observations are 5, 7, 10, 12, 14, 15, then the absolute difference of the remaining two observations is :

(A) 5 (B) 3 (C) 7 (D) 9

Open solution

Q65. JEE Main 2020 (Online) 3rd September Evening Slot

Let x_i $(1 \leq i \leq 10)$ be ten observations of a random variable X. If

$\sum_{i=1}^{10}(x_i - p) = 3$ and $\sum_{i=1}^{10}(x_i - p)^2 = 9$

where $0 \neq p \in R$, then the standard deviation of these observations is :

(A) $\frac{7}{10}$ (B) $\frac{9}{10}$ (C) $\frac{4}{5}$ (D) $\sqrt{\frac{3}{5}}$

Open solution

Q66. JEE Main 2020 (Online) 3rd September Morning Slot

For the frequency distribution :

Variate (x) : x_1 x_2 x_3 .... x_{15}

Frequency (f) : f_1 f_2 f_3 ..... f_{15}

where $0 < x_1 < x_2 < x_3 < ... < x_{15} = 10$ and

$\sum_{i=1}^{15} f_i > 0$, the standard deviation cannot be :

(A) 6 (B) 1 (C) 4 (D) 2

Open solution

Q67. JEE Main 2020 (Online) 2nd September Morning Slot

Let $X = \{x \in N : 1 \leq x \leq 17\}$ and $Y = \{ax + b: x \in X \text{ and } a, b \in R, a > 0\}$. If mean and variance of elements of Y are 17 and 216 respectively then a + b is equal to :

(A) 7 (B) 9 (C) -7 (D) -27

Open solution

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Q68. JEE Main 2020 (Online) 9th January Morning Slot

Let the observations x_i $(1 \leq i \leq 10)$ satisfy the equations, $\sum_{i=1}^{10}(x_1 - 5) = 10$ and $\sum_{i=1}^{10}(x_1 - 5)^2 = 40$.

If μ and λ are the mean and the variance of the observations, $x_1 - 3, x_2 - 3, ..., x_{10} - 3$, then the ordered pair (μ, λ) is equal to :

Open solution

(A) (6, 6) (B) (3, 3) (C) (3, 6) (D) (6, 3)

Q69. JEE Main 2020 (Online) 8th January Evening Slot

The mean and variance of 20 observations are found to be 10 and 4, respectively. On rechecking, it was found that an observation 9 was incorrect and the correct observation was 11. Then the correct variance is

Open solution

(A) 3.98 (B) 3.99 (C) 4.01 (D) 4.02

Q70. JEE Main 2020 (Online) 8th January Morning Slot

The mean and the standard deviation (s.d.) of 10 observations are 20 and 2 resepectively. Each of these 10 observations is multiplied by p and then reduced by q, where $p \neq 0$ and $q \neq 0$. If the new mean and new s.d. become half of their original values, then q is equal to

Open solution

(A) 10 (B) -20 (C) -10 (D) -5

Q71. JEE Main 2019 (Online) 12th April Morning Slot

If the data $x_1, x_2, ..., x_{10}$ is such that the mean of first four of these is 11, the mean of the remaining six is 16 and the sum of squares of all of these is 2,000 ; then the standard deviation of this data is :

Open solution

(A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 4

Q72. JEE Main 2019 (Online) 10th April Evening Slot

If both the mean and the standard deviation of 50 observations $x_1, x_2, ..., x_{50}$ are equal to 16, then the mean of $(x_1 - 4)^2, (x_2 - 4)^2, ..., (x_{50} - 4)^2$ is :

Open solution

(A) 400 (B) 480 (C) 380 (D) 525

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Q73. JEE Main 2019 (Online) 10th April Morning Slot

If for some x ∈ R, the frequency distribution of the marks obtained by 20 students in a test is :

Marks	2	3	5	7
Frequency	$(x+1)^2$	2x - 5	x^2 - 3x	x

then the mean of the marks is

(A) 3.0 (B) 2.8 (C) 2.5 (D) 3.2

Open solution

Q74. JEE Main 2019 (Online) 9th April Evening Slot

The mean and the median of the following ten numbers in increasing order 10, 22, 26, 29, 34, x, 42, 67, 70, y are 42 and 35 respectively, then $\frac{y}{x}$ is equal to

(A) $\frac{7}{2}$ (B) $\frac{8}{3}$ (C) $\frac{9}{4}$ (D) $\frac{7}{3}$

Open solution

Q75. JEE Main 2019 (Online) 9th April Morning Slot

If the standard deviation of the numbers −1, 0, 1, k is $\sqrt{5}$ where k > 0, then k is equal to

(A) $2\sqrt{6}$ (B) $\sqrt{6}$ (C) $2\sqrt{\frac{5}{6}}$ (D) $2\sqrt{\frac{10}{3}}$

Open solution

Q76. JEE Main 2019 (Online) 8th April Evening Slot

A student scores the following marks in five tests :

45, 54, 41, 57, 43.

His score is not known for the sixth test. If the mean score is 48 in the six tests, then the standard deviation of the marks in six tests is

(A) $\frac{100}{\sqrt{3}}$ (B) $\frac{10}{\sqrt{3}}$ (C) $\frac{10}{3}$ (D) $\frac{100}{3}$

Open solution

Q77. JEE Main 2019 (Online) 8th April Morning Slot

The mean and variance of seven observations are 8 and 16, respectively. If 5 of the observations are 2, 4, 10, 12, 14, then the product of the remaining two observations is :

(A) 40 (B) 48 (C) 49 (D) 45

Open solution

Q78. JEE Main 2019 (Online) 12th January Evening Slot

The mean and the variance of five observations are 4 and 5.20, respectively. If three of the observations are 3, 4 and 4 ; then the absolute value of the difference of the other two observations, is :

(A) 1 (B) 7 (C) 3 (D) 5

Open solution

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Q79. JEE Main 2019 (Online) 12th January Morning Slot

If the sum of the deviations of 50 observations from 30 is 50, then the mean of these observations is :

(A) 31 (B) 50 (C) 51 (D) 30

Open solution

Q80. JEE Main 2019 (Online) 11th January Morning Slot

The outcome of each of 30 items was observed; 10 items gave an outcome $\frac{1}{2}$ – d each, 10 items gave outcome $\frac{1}{2}$ each and the remaining 10 items gave outcome $\frac{1}{2}$+ d each. If the variance of this outcome data is $\frac{4}{3}$ then |d| equals :

(A) $\frac{2}{3}$ (B) $\frac{\sqrt{5}}{2}$ (C) $\sqrt{2}$ (D) 2

Open solution

Q81. JEE Main 2019 (Online) 10th January Evening Slot

If mean and standard deviation of 5 observations x_1, x_2, x_3, x_4, x_5 are 10 and 3, respectively, then the variance of 6 observations x_1, x_2, ..., x_5 and −50 is equal to

(A) 582.5 (B) 507.5 (C) 586.5 (D) 509.5

Open solution

Q82. JEE Main 2019 (Online) 10th January Morning Slot

The mean of five observations is 5 and their variance is 9.20. If three of the given five observations are 1, 3 and 8, then a ratio of other two observations is -

(A) 6 : 7 (B) 10 : 3 (C) 4 : 9 (D) 5 : 8

Open solution

Q83. JEE Main 2019 (Online) 9th January Evening Slot

A data consists of n observations : x_1, x_2, ... , x_n.

If $\sum_{i=1}^{n}(x_i+1)^2 = 9n$ and

$\sum_{i=1}^{n}(x_i-1)^2 = 5n$,

then the standard deviation of this data is :

(A) 2 (B) $\sqrt{5}$ (C) 5 (D) $\sqrt{7}$

Open solution

Q84. JEE Main 2019 (Online) 9th January Morning Slot

5 students of a class have an average height 150 cm and variance 18 cm^2. A new student, whose height is 156 cm, joined them. The variance (in cm^2) of the height of these six students is :

(A) 16 (B) 22 (C) 20 (D) 18

Open solution

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Q85. JEE Main 2018 (Online) 16th April Morning Slot

The mean and the standard deviation(s.d.) of five observations are9 and 0, respectively. If one of the observations is changed such that the mean of the new set of five observations becomes 10, then their s.d. is :

Open solution

(A) 0 (B) 1 (C) 2 (D) 4

Q86. JEE Main 2018 (Offline)

If $\sum_{i=1}^{9}(x_i-5)=9$ and

$\sum_{i=1}^{9}(x_i-5)^2=45$, then the standard deviation of the 9 items $x_1, x_2, \ldots\ldots, x_9$ is

Open solution

(A) 3 (B) 9 (C) 4 (D) 2

Q87. JEE Main 2018 (Online) 15th April Evening Slot

If the mean of the data : 7, 8, 9, 7, 8, 7, λ, 8 is 8, then the variance of this data is :

Open solution

(A) $\frac{7}{8}$ (B) 1 (C) $\frac{9}{8}$ (D) 2

Q88. JEE Main 2018 (Online) 15th April Morning Slot

The mean of set of 30 observations is 75. If each observation is multiplied by a non-zero number λ and then each of them is decreased by 25, their mean remains the same. Then λ is equal to :

Open solution

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{10}{3}$

Q89. JEE Main 2017 (Online) 9th April Morning Slot

The sum of 100 observations and the sum of their squares are 400 and 2475, respectively. Later on, three observations, 3, 4 and 5, were found to be incorrect. If the incorrect observations are omitted, then the variance of the remaining observations is :

Open solution

(A) 8.25 (B) 8.50 (C) 8.00 (D) 9.00

Q90. JEE Main 2017 (Online) 8th April Morning Slot

The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If now the mean age of the teachers in this school is 39 years, then the age (in years) of the newly appointed teacher is :

Open solution

(A) 25 (B) 30 (C) 35 (D) 40

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Q91. JEE Main 2016 (Online) 10th April Morning Slot

The mean of 5 observations is 5 and their variance is 12.4. If three of the observations are 1, 2 and 6 ; then the mean deviation from the mean of the data is :

(A) 2.4 (B) 2.8 (C) 2.5 (D) 2.6

Open solution

Q92. JEE Main 2016 (Online) 9th April Morning Slot

If the mean deviation of the numbers 1, 1 + d, ..., 1 +100d from their mean is 255, then a value of d is :

(A) 10.1 (B) 20.2 (C) 10 (D) 5.05

Open solution

Q93. JEE Main 2016 (Offline)

If the standard deviation of the numbers 2, 3, a and 11 is 3.5, then which of the following is true?

(A) $3a^2 - 26a + 55 = 0$ (B) $3a^2 - 32a + 84 = 0$

(C) $3a^2 - 34a + 91 = 0$ (D) $3a^2 - 23a + 44 = 0$

Open solution

Q94. JEE Main 2015 (Online) 10th April Morning Slot

A factory is operating in two shifts, day and night, with 70 and 30 workers respectively. If per day mean wage of the day shift workers is ₹ 54 and per day mean wage of all the workers is ₹ 60, then per day mean wage of the night shift workers (in ₹) is :

(A) 66 (B) 69 (C) 74 (D) 75

Open solution

Q95. JEE Main 2015 (Offline)

The mean of the data set comprising of 16 observations is 16. If one of the observation valued 16 is deleted and three new observations valued 3, 4 and 5 are added to the data, then the mean of the resultant data, is :

(A) 15.8 (B) 14.0 (C) 16.8 (D) 16.0

Open solution

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Q96. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

Let $\overline{x}$, M and σ^2 be respectively the mean, mode and variance of n observations $x_1, x_2, \ldots, x_n$ and $d_i = -x_i - a$, i = 1, 2, ..., n, where a is any number .

Statement I : Variance of $d_1, d_2, \ldots, d_n$ is σ^2.

Statement II : Mean and mode of $d_1, d_2, \ldots, d_n$ are $-\overline{x} - a$ and $-M - a$, respectively

(A) Statement I and Statement II are both false

(B) Statement I and Statement II are both true

(C) Statement I is true and Statement II is false

(D) Statement I is false and Statement II is true

Q97. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

Let $\overline{X}$ and M.D. be the mean and the mean deviation about $\overline{X}$ of n observations x_i, i = 1, 2, ... , n. If each of the observation is increased by 5, then the new mean and the mean deviation about the new mean, respectively, are :

(A) $\overline{X}$, M.D. (B) $\overline{X}$ + 5, M.D.

(C) $\overline{X}$, M.D. + 5 (D) $\overline{X}$ + 5, M.D. + 5

Q98. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

In a set of 2n distinct observations, each of the observation below the median of all the observations is increased by 5 and each of the remaining observations is decreased by 3. Then the mean of the new set of observations :

(A) increases by 1. (B) decreases by 1. (C) decreases by 2. (D) increases by 2.

Q99. JEE Main 2014 (Offline)

Open solution

The variance of first 50 even natural numbers is

(A) 833 (B) 437 (C) $\frac{437}{4}$ (D) $\frac{833}{4}$

Q100. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

In a set of 2n observations, half of them are equal to 'a' and the remaining half are equal to '-a'. If the standard deviation of all the observations is 2; then the value of $|a|$ is

(A) 2 (B) $\sqrt{2}$ (C) 4 (D) $2\sqrt{2}$

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Q101. JEE Main 2013 (Online) 23th April Morning Slot

If the median and the range of four numbers {x, y, 2x + y, x - y}, where 0 < y < x < 2y, are 10 and 28 respectively, then the mean of the numbers is:

(A) 18 (B) 10 (C) 5 (D) 14

Open solution

Q102. JEE Main 2013 (Online) 22th April Morning Slot

Mean of 5 observations is 7. If four of those observations are 6, 7, 8, 10 and one is missing then the variance of all the five observations is:

(A) 8 (B) 4 (C) 6 (D) 2

Open solution

Q103. JEE Main 2013 (Online) 9th April Morning Slot

The mean of a data set consisting of 20 observations is 40. If one observation 53 was wrongly recorded as 33, then the correct mean will be:

(A) 41 (B) 49 (C) 40.5 (D) 42.5

Open solution

Q104. JEE Main 2013 (Offline)

All the students of a class performed poorly in Mathematics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given?

(A) median (B) mode (C) variance (D) mean

Open solution

Q105. AIEEE 2012 (Online) 26th May

Statement 1: The variance of first n odd natural numbers is $\frac{n^2-1}{3}$
Statement 2: The sum of first n odd natural number is n^2 and the sum of square of first n odd natural numbers is $\frac{n(4n^2+1)}{3}$.

(A) Statement 1 is true, Statement 2 is false.

(B) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.

(C) Statement 1 is false, Statement 2 is true.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation for Statement 1.

Open solution

Q106. AIEEE 2012 (Online) 19th May

The median of 100 observations grouped in classes of equal width is 25. If the median class interval is 20-30 and the number of observations less than 20 is 45, then the frequency of median class is

(A) 10 (B) 20 (C) 15 (D) 12

Open solution

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Q107. AIEEE 2012 (Online) 12th May

If the mean of $4, 7, 2, 8, 6$ and a is 7, then the mean deviation from the median of these observations is

(A) 8 (B) 5 (C) 1 (D) 3

Open solution

Q108. AIEEE 2012 (Online) 7th May

The frequency distribution of daily working expenditure of families in a locality is as follows:

Expenditure (in Rs.) (x)	0-50	50-100	100-150	150-200	200-250
No. of families (f)	24	33	37	b	25

If the mode of the distribution is Rs. 140, then the value of b is

(A) 34 (B) 31 (C) 26 (D) 36

Open solution

Q109. AIEEE 2012

Let $x_1, x_2, \ldots\ldots, x_n$ be n observations, and let $\overline{x}$ be their arithematic mean and σ^2 be their variance.

Statement 1 : Variance of $2x_1, 2x_2, \ldots\ldots, 2x_n$ is $4\sigma^2$.

Statement 2 : : Arithmetic mean of $2x_1, 2x_2, \ldots\ldots, 2x_n$ is $4\overline{x}$.

(A) Statement 1 is false, statement 2 is true

(B) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1

(C) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1

(D) Statement 1 is true, statement 2 is false

Open solution

Q110. AIEEE 2011

If the mean deviation about the median of the numbers a, 2a,....., 50a is 50, then |a| equals

(A) 4 (B) 5 (C) 2 (D) 3

Open solution

Q111. AIEEE 2010

For two data sets, each of size 5, the variances are given to be 4 and 5 and the corresponding means are given to be 2 and 4, respectively. The variance of the combined data set is

(A) $\frac{5}{2}$ (B) $\frac{11}{2}$ (C) 6 (D) $\frac{13}{2}$

Open solution

Q112. AIEEE 2009

If the mean deviation of numbers 1, 1 + d, 1 + 2d,....., 1 + 100d from their mean is 255, then d is equal to

(A) 20.0 (B) 10.1 (C) 20.2 (D) 10.0

Open solution

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Q113. AIEEE 2009

Statement - 1 : The variance of first n even natural numbers is $\frac{n^2-1}{4}$

Statement - 2 : The sum of first n natural numbers is $\frac{n(n+1)}{2}$ and the sum of squares of first n natural numbers is $\frac{n(n+1)(2n+1)}{6}$

Open solution

(A) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1

(C) Statement-1 is true, Statement-2 is false

(D) Statement-1 is false, Statement-2 is true

Q114. AIEEE 2008

The mean of the numbers a, b, 8, 5, 10 is 6 and the variance is 6.80. Then which one of the following gives possible values of a and b?

Open solution

(A) a = 0, b = 7 (B) a = 5, b = 2 (C) a = 1, b = 6 (D) a = 3, b = 4

Q115. AIEEE 2007

The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is

Open solution

(A) 80 (B) 60 (C) 40 (D) 20

Q116. AIEEE 2006

Suppose a population A has 100 observations 101, 102,....., 200, and another population B has 100 observations 151, 152,....., 250. If V_A and V_B represent the variances of the two populations, respectively, then $\frac{V_A}{V_B}$ is

Open solution

(A) 1 (B) $\frac{9}{4}$ (C) $\frac{4}{9}$ (D) $\frac{2}{3}$

Q117. AIEEE 2005

If in a frequency distribution, the mean and median are 21 and 22 respectively, then its mode is approximately :

Open solution

(A) 20.5 (B) 22.0 (C) 24.0 (D) 25.5

Q118. AIEEE 2005

Let $x_1, x_2, \ldots\ldots\ldots, x_n$ be n observations such that

$\sum x_i^2 = 400$ and $\sum x_i = 80$. Then a possible value of n among the following is

Open solution

(A) 18 (B) 15 (C) 12 (D) 9

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Q119. AIEEE 2004

Consider the following statements:
(a) Mode can be computed from histogram
(b) Median is not independent of change of scale
(c) Variance is independent of change of origin and scale.
Which of these is/are correct?

(A) only (a) (B) only (b) (C) only (a) and (b) (D) (a), (b) and (c)

Open solution

Q120. AIEEE 2004

In a series of 2n observations, half of them equal a and remaining half equal $-a$. If the standard deviation of the observations is 2, then $|a|$ equals

(A) 2 (B) $\sqrt{2}$ (C) $\frac{1}{n}$ (D) $\frac{\sqrt{2}}{n}$

Open solution

Q121. AIEEE 2003

In an experiment with 15 observations on x, the following results were available:
$\sum x^2 = 2830, \sum x = 170$
One observation that was 20 was found to be wrong and was replaced by the correct value 30. Then the corrected variance is :

(A) 188.66 (B) 177.33 (C) 8.33 (D) 78.00

Open solution

Q122. AIEEE 2003

The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set :

(A) is increased by 2
(B) is decreased by 2
(C) is two times the original median
(D) remains the same as that of the original set

Open solution

Q123. AIEEE 2002

In a class of 100 students there are 70 boys whose average marks in a subject are 75. If the average marks of the complete class is 72, then what is the average marks of the girls?

(A) 73 (B) 65 (C) 68 (D) 74

Open solution

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Numerical 38 questions

Q124. JEE Main 2025 (Online) 23rd January Evening Shift

The variance of the numbers $8, 21, 34, 47, \ldots, 320$ is _____.

Open solution

Q125. JEE Main 2024 (Online) 8th April Evening Shift

Let $\mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathbf{N}$ and $\mathrm{a} < \mathrm{b} < \mathrm{c}$. Let the mean, the mean deviation about the mean and the variance of the 5 observations $9, 25, a, b, c$ be 18, 4 and $\frac{136}{5}$, respectively. Then $2a + b - c$ is equal to _____

Open solution

Q126. JEE Main 2024 (Online) 5th April Evening Shift

Let the mean and the standard deviation of the probability distribution

X	α	1	0	-3
P(X)	$\frac{1}{3}$	K	$\frac{1}{6}$	$\frac{1}{4}$

be μ and σ, respectively. If $\sigma - \mu = 2$, then $\sigma + \mu$ is equal to _____.

Open solution

Q127. JEE Main 2024 (Online) 30th January Evening Shift

The variance σ^2 of the data

x_i	0	1	5	6	10	12	17
f_i	3	2	3	2	6	3	3

is _____.

Open solution

Q128. JEE Main 2024 (Online) 29th January Morning Shift

If the mean and variance of the data $65, 68, 58, 44, 48, 45, 60, \alpha, \beta, 60$ where $\alpha > \beta$, are 56 and 66.2 respectively, then $\alpha^2 + \beta^2$ is equal to _____.

Open solution

Q129. JEE Main 2024 (Online) 27th January Evening Shift

The mean and standard deviation of 15 observations were found to be 12 and 3 respectively. On rechecking it was found that an observation was read as 10 in place of 12 . If μ and σ^2 denote the mean and variance of the correct observations respectively, then $15\left(\mu + \mu^2 + \sigma^2\right)$ is equal to _____.

Open solution

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Q130. JEE Main 2023 (Online) 13th April Evening Shift

The mean and standard deviation of the marks of 10 students were found to be 50 and 12 respectively. Later, it was observed that two marks 20 and 25 were wrongly read as 45 and 50 respectively. Then the correct variance is _____

Open solution

Q131. JEE Main 2023 (Online) 13th April Morning Shift

Let the mean of the data

x	1	3	5	7	9
Frequency (f)	4	24	28	α	8

be 5. If m and σ^2 are respectively the mean deviation about the mean and the variance of the data, then $\frac{3\alpha}{m+\sigma^2}$ is equal to _____

Open solution

Q132. JEE Main 2023 (Online) 12th April Morning Shift

Let the positive numbers a_1, a_2, a_3, a_4 and a_5 be in a G.P. Let their mean and variance be $\frac{31}{10}$ and $\frac{m}{n}$ respectively, where m and n are co-prime. If the mean of their reciprocals is $\frac{31}{40}$ and $a_3 + a_4 + a_5 = 14$, then $m + n$ is equal to _____.

Open solution

Q133. JEE Main 2023 (Online) 10th April Morning Shift

If the mean of the frequency distribution

Class :	0-10	10-20	20-30	30-40	40-50
Frequency :	2	3	x	5	4

is 28, then its variance is _____.

Open solution

Q134. JEE Main 2023 (Online) 8th April Morning Shift

Let the mean and variance of 8 numbers $x, y, 10, 12, 6, 12, 4, 8$ be 9 and 9.25 respectively. If $x > y$, then $3x - 2y$ is equal to _____.

Open solution

Q135. JEE Main 2023 (Online) 6th April Evening Shift

If the mean and variance of the frequency distribution

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

are 9 and 15.08 respectively, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is _____.

Open solution

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Q136. JEE Main 2023 (Online) 31st January Morning Shift

If the variance of the frequency distribution

x_i	2	3	4	5	6	7	8
Frequency f_i	3	6	16	α	9	5	6

is 3, then α is equal to _____.

Open solution

Q137. JEE Main 2023 (Online) 30th January Morning Shift

The mean and variance of 7 observations are 8 and 16 respectively. If one observation 14 is omitted and a and b are respectively mean and variance of remaining 6 observations, then $\mathrm{a}+3\mathrm{b}-5$ is equal to _____.

Open solution

Q138. JEE Main 2023 (Online) 29th January Evening Shift

Let $X=\{11,12,13,\ldots,40,41\}$ and $Y=\{61,62,63,\ldots,90,91\}$ be the two sets of observations. If $\overline{x}$ and $\overline{y}$ are their respective means and σ^2 is the variance of all the observations in $\mathrm{X}\cup\mathrm{Y}$, then $\left|\overline{x}+\overline{y}-\sigma^2\right|$ is equal to _____.

Open solution

Q139. JEE Main 2022 (Online) 29th July Morning Shift

Let the mean and the variance of 20 observations $x_1, x_2, \ldots, x_{20}$ be 15 and 9 , respectively. For $\alpha \in \mathbf{R}$, if the mean of $(x_1+\alpha)^2,(x_2+\alpha)^2,\ldots,(x_{20}+\alpha)^2$ is 178 , then the square of the maximum value of α is equal to _____.

Open solution

Q140. JEE Main 2022 (Online) 28th July Morning Shift

Let $x_1, x_2, x_3, \ldots, x_{20}$ be in geometric progression with $x_1=3$ and the common ratio $\frac{1}{2}$. A new data is constructed replacing each x_i by $(x_i-i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal to $\bar{x}$ is _____.

Open solution

Q141. JEE Main 2022 (Online) 27th July Morning Shift

The mean and variance of 10 observations were calculated as 15 and 15 respectively by a student who took by mistake 25 instead of 15 for one observation. Then, the correct standard deviation is _____.

Open solution

Q142. JEE Main 2022 (Online) 26th July Evening Shift

The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of these observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to _____.

Open solution

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Q143. JEE Main 2022 (Online) 28th June Evening Shift

Suppose a class has 7 students. The average marks of these students in the mathematics examination is 62, and their variance is 20. A student fails in the examination if he/she gets less than 50 marks, then in worst case, the number of students can fail is _____.

Open solution

Q144. JEE Main 2022 (Online) 28th June Morning Shift

The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5. Then, the correct variance is equal to _____.

Open solution

Q145. JEE Main 2022 (Online) 25th June Evening Shift

If the mean deviation about the mean of the numbers 1, 2, 3, ..., n, where n is odd, is $\frac{5(n+1)}{n}$, then n is equal to _____.

Open solution

Q146. JEE Main 2021 (Online) 31st August Morning Shift

The mean of 10 numbers 7×8, 10×10, 13×12, 16×14, ... is _____.

Open solution

Q147. JEE Main 2021 (Online) 27th August Evening Shift

An online exam is attempted by 50 candidates out of which 20 are boys. The average marks obtained by boys is 12 with a variance 2. The variance of marks obtained by 30 girls is also 2. The average marks of all 50 candidates is 15. If μ is the average marks of girls and σ^2 is the variance of marks of 50 candidates, then $\mu + \sigma^2$ is equal to _____.

Open solution

Q148. JEE Main 2021 (Online) 27th August Morning Shift

Let n be an odd natural number such that the variance of 1, 2, 3, 4, ..., n is 14. Then n is equal to _____.

Open solution

Q149. JEE Main 2021 (Online) 26th August Evening Shift

Let the mean and variance of four numbers 3, 7, x and y(x > y) be 5 and 10 respectively. Then the mean of four numbers 3 + 2x, 7 + 2y, x + y and $x - y$ is _____.

Open solution

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Q150. JEE Main 2021 (Online) 25th July Morning Shift

Consider the following frequency distribution :

Class :	10-20	20-30	30-40	40-50	50-60
Frequency :	α	110	54	30	β

If the sum of all frequencies is 584 and median is 45, then | $\alpha - \beta$ | is equal to _____.

Open solution

Q151. JEE Main 2021 (Online) 22th July Evening Shift

Consider the following frequency distribution :

Class :	0-6	6-12	12-18	18-24	24-30
Frequency :	a	b	12	9	5

If mean = $\frac{309}{22}$ and median = 14, then the value $(a - b)^2$ is equal to _____.

Open solution

Q152. JEE Main 2021 (Online) 18th March Morning Shift

The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If the mean age of the teachers in this school now is 39 years, then the age (in years) of the newly appointed teacher is _____.

Open solution

Q153. JEE Main 2021 (Online) 17th March Evening Shift

Consider a set of 3n numbers having variance 4. In this set, the mean of first 2n numbers is 6 and the mean of the remaining n numbers is 3. A new set is constructed by adding 1 into each of first 2n numbers, and subtracting 1 from each of the remaining n numbers. If the variance of the new set is k, then 9k is equal to _____.

Open solution

Q154. JEE Main 2021 (Online) 16th March Evening Shift

Consider the statistics of two sets of observations as follows :

	Size	Mean	Variance
Observation I	10	2	2
Observation II	n	3	1

If the variance of the combined set of these two observations is $\frac{17}{9}$, then the value of n is equal to _____.

Open solution

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Q155. JEE Main 2021 (Online) 26th February Evening Shift

Let $X_1, X_2, \ldots\ldots, X_{18}$ be eighteen observations such that $\sum_{i=1}^{18}(X_i-\alpha)=36$ and $\sum_{i=1}^{18}(X_i-\beta)^2=90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha-\beta|$ is _____.

Open solution

Q156. JEE Main 2021 (Online) 24th February Evening Shift

If the variance of 10 natural numbers 1, 1, 1, ..., 1, k is less than 10, then the maximum possible value of k is _____.

Open solution

Q157. JEE Main 2020 (Online) 6th September Evening Slot

Consider the data on x taking the values 0, 2, 4, 8,....., 2^n with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, \ldots., {}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$, then n is equal to _____ .

Open solution

Q158. JEE Main 2020 (Online) 4th September Evening Slot

If the variance of the following frequency distribution :

Class	10–20	20–30	30–40
Frequency :	2	x	2

is 50, then x is equal to____

Open solution

Q159. JEE Main 2020 (Online) 2nd September Evening Slot

If the variance of the terms in an increasing A.P., $b_1, b_2, b_3, \ldots., b_{11}$ is 90, then the common difference of this A.P. is_____.

Open solution

Q160. JEE Main 2020 (Online) 7th January Evening Slot

If the mean and variance of eight numbers 3, 7, 9, 12, 13, 20, x and y be 10 and 25 respectively, then x.y is equal to _____.

Open solution

Q161. JEE Main 2020 (Online) 7th January Morning Slot

If the variance of the first n natural numbers is 10 and the variance of the first m even natural numbers is 16, then m + n is equal to_____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **B**	3. **B**	4. **A**	5. **C**
6. **D**	7. **D**	8. **D**	9. **C**	10. **C**
11. **D**	12. **D**	13. **B**	14. **A**	15. **D**
16. **C**	17. **C**	18. **B**	19. **A**	20. **A**
21. **B**	22. **B**	23. **D**	24. **C**	25. **B**
26. **D**	27. **A**	28. **C**	29. **A**	30. **C**
31. **A**	32. **D**	33. **C**	34. **B**	35. **C**
36. **B**	37. **A**	38. **C**	39. **D**	40. **D**
41. **D**	42. **D**	43. **C**	44. **B**	45. **A**
46. **D**	47. **A**	48. **B**	49. **B**	50. **C**
51. **A**	52. **C**	53. **D**	54. **C**	55. **D**
56. **C**	57. **D**	58. **A**	59. **A**	60. **B**
61. **D**	62. **D**	63. **A**	64. **C**	65. **B**
66. **A**	67. **C**	68. **B**	69. **B**	70. **B**
71. **B**	72. **A**	73. **B**	74. **D**	75. **A**
76. **B**	77. **B**	78. **B**	79. **A**	80. **C**
81. **B**	82. **C**	83. **B**	84. **C**	85. **C**
86. **D**	87. **B**	88. **C**	89. **D**	90. **C**
91. **B**	92. **A**	93. **B**	94. **C**	95. **B**
96. **B**	97. **B**	98. **A**	99. **A**	100. **A**
101. **D**	102. **B**	103. **A**	104. **C**	105. **A**
106. **A**	107. **D**	108. **D**	109. **D**	110. **A**
111. **B**	112. **B**	113. **D**	114. **D**	115. **A**
116. **A**	117. **C**	118. **A**	119. **C**	120. **A**
121. **D**	122. **D**	123. **B**		

Numerical

124. **8788**	125. **33**	126. **5**	127. **29**	128. **6344**
129. **2521**	130. **269**	131. **8**	132. **211**	133. **151**
134. **25**	135. **25**	136. **5**	137. **37**	138. **603**
139. **4**	140. **142**	141. **2**	142. **238**	143. **0**

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144. **17**	145. **21**	146. **398**	147. **25**	148. **13**
149. **12**	150. **164**	151. **4**	152. **35**	153. **68**
154. **5**	155. **4**	156. **11**	157. **6**	158. **4**
159. **3**	160. **54**	161. **18**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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